EE2004: Digital Signal Processing

Tutorial 1 Answer Key

1. (a) energy signal, (b) power signal, (c) power signal, (d) neither an energy signal nor a power signal.

2. (a) $\{\overset{\downarrow}{0}, 0, 5, 2, 1, 0, -1\}$, (b) $x[n]$, (c) $5\delta[n-3]$, (d) $\{2, \overset{\downarrow}{1}, 2\}$, (e) $2\delta[n]$

3. (a) $n < -6$ and $n > 0$, (b) $n < -4$ and $n > 2$, and (c) $n < -2$ and $n > 4$

4. Proof

5. (a) $x_e[n] =$ plot of $0.5\,(u[n+5] - u[n-6]) + 0.5\,\delta[n]$
   $x_o[n] =$ plot of $0.5\,(u[n-1] - u[n-6]) - 0.5\,(u[n+5] - u[n])$
   (b) $x_e[n] =$ plot of $0.5\,(\delta[n+1] + \delta[n-1] + \delta[n+3] + \delta[n-3]) + (\delta[n+2] + \delta[n-2])$
   $x_o[n] =$ plot of $0.5\,(\delta[n-1] - \delta[n+1] + \delta[n-3] - \delta[n+3]) + (\delta[n-2] - \delta[n+2])$

6. (a) $x_e[n] = (1 + \delta[n])/2$, $x_o[n] = 0.5\,\mathrm{sgn}[n]$, (b) $x_e[n] = \frac{|n|}{2}$, $x_o[n] = \frac{n}{2}$, (c) $x_e[n] = 0$, $x_o[n] = \sin(\frac{n\pi}{4})$, (d) $x_e[n] = \cos(\frac{n\pi}{4})$, $x_o[n] = 0$, (e) $x_e[n] = 0.5\,\mathrm{sgn}[n]\,\sin(\frac{n\pi}{4})$, $x_o[n] = 0.5\sin(\frac{n\pi}{4})$, (f) $x_e[n] = 0.5\,\big(\cos(\frac{n\pi}{4}) + \delta[n]\big)$, $x_o[n] = 0.5\,\big(\cos(\frac{n\pi}{4})\,\mathrm{sgn}[n] - \delta[n]\big)$, (g) $x_e[n] = \cos(\frac{n\pi}{4})\cos(\frac{\pi}{4})$, $x_o[n] = -\sin(\frac{n\pi}{4})\sin(\frac{\pi}{4})$, (h) $x_e[n] = \dfrac{\delta[n+5] + \delta[n-5]}{2} + u[n+4] - u[n-5]$, $x_o[n] = \dfrac{\delta[n+5] - \delta[n-5]}{2}$

7. $x_{ce}[n] = e^{jn} = x[n]$ and $x_{co}[n] = 0$

8. 0 (under what conditions is this answer true?)

9. (a) $z_0 = e^{-1}$, (b) $c = \ln 2$

10. (a) 0, (b) $-\delta(t - \pi)$.